

Azure Functions

Serverless Compute on Microsoft Azure

1. What is Azure Functions?

Azure Functions is a serverless compute service in Microsoft Azure that allows developers to run small pieces of code without managing the underlying server infrastructure. Functions are commonly used for event-driven applications where code executes in response to events such as HTTP requests, timers, messages, and changes in Azure Storage.

2. Triggers

A trigger defines the event that causes an Azure Function to execute. Common triggers include HTTP triggers, Timer triggers, Azure Blob Storage triggers, Azure Queue Storage triggers, and Azure Service Bus triggers. An HTTP trigger can expose a function through a REST endpoint.

3. Bindings

Bindings provide a way for an Azure Function to connect to other Azure services without writing extensive connection code. Input bindings provide data to a function, while output bindings send data to another service. Examples include bindings for Blob Storage, Queue Storage, Cosmos DB, and Service Bus.

4. Hosting Plans

Azure Functions can run using different hosting options. Consumption plans automatically scale based on demand and charge according to execution and resource usage. Premium hosting provides additional features such as pre-warmed instances and more predictable performance. Dedicated hosting runs functions on dedicated App Service infrastructure.

5. Durable Functions

Durable Functions is an extension of Azure Functions that helps developers implement stateful workflows in a serverless environment. It supports orchestrator functions, activity functions, and durable entities. Durable Functions can be used for long-running processes, function chaining, fan-out and fan-in patterns, and human interaction workflows.

6. Common Use Cases

Common Azure Functions scenarios include REST APIs, scheduled jobs, file processing, message processing, event-driven automation, background tasks, data transformation, and integration with Azure services such as Event Grid, Service Bus, Blob Storage, and Cosmos DB.